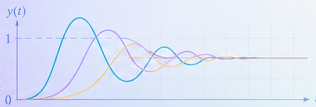
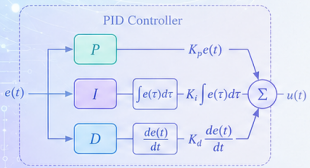
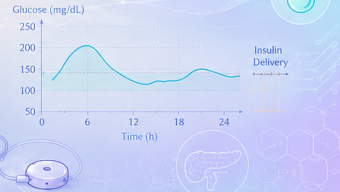
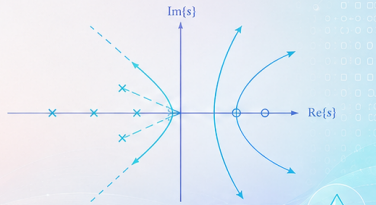
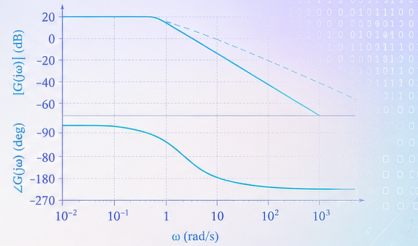


Steady-State Errors

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$$G(s) = \frac{Y(s)}{U(s)} \quad a_n \frac{d^n y(t)}{dt^n} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \dots + b_m \frac{d^m u(t)}{dt^m} + \dots + b_1 \frac{du(t)}{dt} + b_0 u(t)$$



Steady-State Error

- ▶ The **steady-state error** is the difference between the desired input and the actual output as $t \rightarrow \infty$.
- ▶ If the error signal is

$$e(t) = r(t) - c(t),$$

then the steady-state error is defined as

$$e_{ss} = \lim_{t \rightarrow \infty} e(t).$$

- ▶ Ideally,

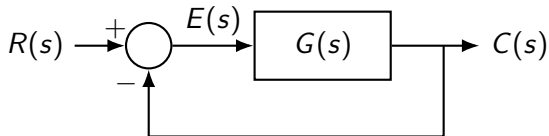
$$e_{ss} = 0$$

which means that the output eventually reaches the desired input.

Steady-State Error

For a prescribed test input,

$$e_{ss} = \lim_{t \rightarrow \infty} [r(t) - c(t)]$$

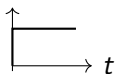


Standard Test Inputs for Steady-State Error

Standard test inputs are used to evaluate and compare the **steady-state accuracy**.

Waveform	Name	Physical Interpretation	Time Function	Laplace Transform
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$r(t)$



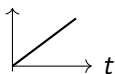
Step

Constant position

1

$\frac{1}{s}$

$r(t)$



Ramp

Constant velocity

t

$\frac{1}{s^2}$

$r(t)$



Parabola

Constant acceleration

$\frac{1}{2}t^2$

$\frac{1}{s^3}$

Steady-State Error for Unity Feedback Systems

For a unity-feedback system, steady-state error can be evaluated using either

closed-loop transfer function $T(s)$

or

open-loop transfer function $G(s)$.

Using $T(s)$

$$C(s) = T(s)R(s)$$

and

$$E(s) = R(s) - C(s).$$

Using $G(s)$

For unity feedback,

$$C(s) = G(s)E(s).$$

This form is especially useful for identifying the factors that affect steady-state error.

Steady-State Error in Terms of $T(s)$

For a unity-feedback system,

$$E(s) = R(s) - C(s).$$

Since

$$C(s) = R(s)T(s),$$

substituting gives

$$\begin{aligned} E(s) &= R(s) - R(s)T(s) \\ &= R(s)[1 - T(s)]. \end{aligned}$$

Therefore,

Error in Terms of Closed-Loop Transfer Function

$$E(s) = R(s)[1 - T(s)]$$

Steady-State Error Using Final Value Theorem

The steady-state error is

$$e_{ss} = e(\infty) = \lim_{t \rightarrow \infty} e(t).$$

Using the **final value theorem**,

$$e(\infty) = \lim_{s \rightarrow 0} sE(s).$$

Since

$$E(s) = R(s)[1 - T(s)],$$

we obtain

$$e_{ss} = \lim_{s \rightarrow 0} sR(s)[1 - T(s)]$$

Important

The final value theorem can be applied only when the corresponding closed-loop system satisfies the required stability conditions.

Steady-State Error in Terms of $G(s)$

For a unity-feedback system,

$$E(s) = R(s) - C(s).$$

Also,

$$C(s) = E(s)G(s).$$

Substituting,

$$E(s) = R(s) - E(s)G(s).$$

Hence,

$$E(s)[1 + G(s)] = R(s),$$

Error Transfer Relation

$$E(s) = \frac{R(s)}{1 + G(s)}$$

General Steady-State Error Formula

Using

$$E(s) = \frac{R(s)}{1 + G(s)}$$

and the final value theorem,

$$e_{ss} = \lim_{s \rightarrow 0} sE(s).$$

Therefore,

$$e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)}$$

- ▶ This expression relates steady-state error directly to the **open-loop transfer function** $G(s)$.
- ▶ The nature of e_{ss} depends strongly on the behaviour of $G(s)$ as

$$s \rightarrow 0.$$

Steady-State Error for a Step Input

For a unit-step input,

$$R(s) = \frac{1}{s}.$$

Using

$$e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)},$$

we obtain

$$e_{\text{step}} = \lim_{s \rightarrow 0} \frac{s(1/s)}{1 + G(s)} = \frac{1}{1 + \lim_{s \rightarrow 0} G(s)}.$$

Define the **position error constant**

$$K_p = \lim_{s \rightarrow 0} G(s).$$

Hence

$$e_{\text{step}} = \frac{1}{1 + K_p}$$

Step Input: Condition for Zero Error

For

$$e_{\text{step}} = \frac{1}{1 + \lim_{s \rightarrow 0} G(s)},$$

zero steady-state error requires

$$\lim_{s \rightarrow 0} G(s) = \infty$$

If the forward path contains at least one integrator,

$$G(s) = \frac{K}{s} \frac{\text{other factors}}{\text{other factors}},$$

then $G(s) \rightarrow \infty$ ($s \rightarrow 0$).

Therefore,

$$\text{At least one integrator} \Rightarrow e_{\text{step}} = 0$$

An integrator corresponds to a pole at

$$s = 0.$$

Steady-State Error for a Ramp Input

For a unit-ramp input,

$$R(s) = \frac{1}{s^2}.$$

Therefore,

$$\begin{aligned} e_{\text{ramp}} &= \lim_{s \rightarrow 0} \frac{s(1/s^2)}{1 + G(s)} \\ &= \lim_{s \rightarrow 0} \frac{1}{s + sG(s)} = \frac{1}{\lim_{s \rightarrow 0} sG(s)}. \end{aligned}$$

Define the **velocity error constant**

$$K_v = \lim_{s \rightarrow 0} sG(s).$$

Hence,

$$e_{\text{ramp}} = \frac{1}{K_v}$$

Ramp Input: Effect of Integrators

For a ramp input,

$$e_{\text{ramp}} = \frac{1}{\lim_{s \rightarrow 0} sG(s)}.$$

Two or More Integrators

If $G(s)$ contains at least two factors $1/s$,
 $\lim_{s \rightarrow 0} sG(s) = \infty$, therefore,

$$e_{\text{ramp}} = 0.$$

One Integrator

If $G(s)$ contains one integrator,
 $\lim_{s \rightarrow 0} sG(s) = \text{finite}$, so the ramp error
is a **finite constant**.

No Integrator

If $G(s)$ contains no integrator,

$$\lim_{s \rightarrow 0} sG(s) = 0.$$

Therefore, $e_{\text{ramp}} = \infty$ and the output
cannot track the ramp with a bounded
steady-state error.

Steady-State Error for a Parabolic Input

For a unit-parabolic input,

$$R(s) = \frac{1}{s^3}.$$

Hence,

$$\begin{aligned} e_{\text{parabolic}} &= \lim_{s \rightarrow 0} \frac{s(1/s^3)}{1 + G(s)} \\ &= \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 G(s)} = \frac{1}{\lim_{s \rightarrow 0} s^2 G(s)}. \end{aligned}$$

Define the **acceleration error constant**

$$K_a = \lim_{s \rightarrow 0} s^2 G(s).$$

Therefore,

$$e_{\text{parabolic}} = \frac{1}{K_a}$$

Parabolic Input: Effect of Integrators

For a parabolic input,

$$e_{\text{parabolic}} = \frac{1}{\lim_{s \rightarrow 0} s^2 G(s)}.$$

Three or More Integrators

$$\lim_{s \rightarrow 0} s^2 G(s) = \infty$$

and therefore, $e_{\text{parabolic}} = 0$

Two Integrators

$\lim_{s \rightarrow 0} s^2 G(s) = \text{finite}$. Hence, $e_{\text{parabolic}}$ is a **finite constant**.

One or No Integrator

$\lim_{s \rightarrow 0} s^2 G(s) = 0$. Therefore,

$$e_{\text{parabolic}} = \infty$$

Thus, one or no integration in the forward path cannot provide bounded steady-state tracking of a parabolic input.

Static Error Constants

For unity-feedback systems, define

$$\boxed{K_p = \lim_{s \rightarrow 0} G(s)} \quad \boxed{K_v = \lim_{s \rightarrow 0} sG(s)} \quad \boxed{K_a = \lim_{s \rightarrow 0} s^2 G(s)}$$

Input	Error Constant	Steady-State Error
Step	K_p	$\frac{1}{1 + K_p}$
Ramp	K_v	$\frac{1}{K_v}$
Parabola	K_a	$\frac{1}{K_a}$

Number of Integrators and Tracking Accuracy

Suppose the forward transfer function contains N poles at the origin:

$$G(s) = \frac{K N(s)}{s^N D(s)},$$

where $N(0) \neq 0$ and $D(0) \neq 0$.

No. of Integrators	Step	Ramp	Parabola
$N = 0$	Finite	∞	∞
$N = 1$	0	Finite	∞
$N = 2$	0	0	Finite
$N \geq 3$	0	0	0

Steady-State Error: Key Relationships

For a unity-feedback system,

$$e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)}$$

Step: $R(s) = \frac{1}{s}$

$$e_{ss} = \frac{1}{1 + K_p}$$

Ramp: $R(s) = \frac{1}{s^2}$

$$e_{ss} = \frac{1}{K_v}$$

Parabola: $R(s) = \frac{1}{s^3}$

$$e_{ss} = \frac{1}{K_a}$$

where

$$K_p = \lim_{s \rightarrow 0} G(s),$$

$$K_v = \lim_{s \rightarrow 0} sG(s), \quad K_a = \lim_{s \rightarrow 0} s^2 G(s).$$

Steady-State Error for Systems in State Space

Consider the closed-loop SISO system

$$\dot{\mathbf{x}} = A\mathbf{x} + B r,$$

$$y = C\mathbf{x}.$$

The corresponding closed-loop transfer function is

$$T(s) = C(sI - A)^{-1}B$$

Two approaches can be used to determine steady-state error:

1. Final Value Theorem
2. Input Substitution

We first consider the final-value-theorem approach.

Objective

Determine

$$e_{ss} = \lim_{t \rightarrow \infty} e(t)$$

directly from the state-space model.

Steady-State Error via Final Value Theorem

The error between the reference input and the output is

$$E(s) = R(s) - Y(s).$$

Since

$$Y(s) = R(s)T(s),$$

we obtain

$$\begin{aligned} E(s) &= R(s) - R(s)T(s) \\ &= R(s)[1 - T(s)]. \end{aligned}$$

Therefore,

Error in Terms of the Closed-Loop Transfer Function

$$E(s) = R(s)[1 - T(s)]$$

Steady-State Error in Terms of State-Space Matrices

For the state-space system,

$$T(s) = C(sI - A)^{-1}B.$$

Substituting this into

$$E(s) = R(s) [1 - T(s)],$$

gives

$$E(s) = R(s) [1 - C(sI - A)^{-1}B]$$

Steady-State Error in Terms of State-Space Matrices

For the state-space system,

$$T(s) = C(sI - A)^{-1}B.$$

Substituting this into

$$E(s) = R(s) [1 - T(s)],$$

gives

$$E(s) = R(s) [1 - C(sI - A)^{-1}B]$$

Using the **final value theorem**,

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s).$$

Steady-State Error from State-Space Model

$$e_{ss} = \lim_{s \rightarrow 0} sR(s) [1 - C(sI - A)^{-1}B]$$

Thank You

Questions and Discussion